Does the Narrative See-Saw of Monetary Policy Announcements Affect Market Expectations?

Paul L. Tran*

This version: 23 May, 2026

[Click here for the latest version]

Abstract

This paper shows that the thematic composition of FOMC statements affects market expectations of future monetary policy. I quantify changes in communication bandwidth across topics using large-language-model probabilistic topic classification. Using an event-study approach, I document a "narrative see-saw" effect: a relative expansion of monetary policy tool discussions, which compresses macroeconomic context, amplifies the variation of monetary policy surprises. These signals remain significant after controlling for policy actions and semantic novelty. Including both textual measures improves explanatory power by 30% relative to the policy action alone. Thus, markets respond to the simultaneous trade-off between policy discussion and macroeconomic context.

Keywords: FOMC statements, monetary policy surprises, text analysis, probabilistic topic classification, large language models, event window studies

JEL Codes: C45, E52, E58, G14

*Office of Macroeconomic Analysis, US Department of the Treasury (incoming). Department of Economics, The University of Texas at Austin. 2225 Speedway, BRB 2.128, C3100, Austin, TX 78712, USA. Email: pltran@utexas.edu. Website: https://paulletran.com/. I am indebted to my committee—Oli Coibion, Chris Boehm, Saroj Bhattarai, and Amy Handlan—for their invaluable guidance and support. For their helpful comments and suggestions, I also wish to thank Oliver Pfauti, Nasiha Salwait, and Adry Gracio. I am grateful to Chris Boehm and Niklas Kroner for sharing data with me. I also appreciate the Empirical Macroeconomics Policy Center of Texas for their financial support to cover some of the computational costs behind my paper. Views expressed are my own and do not necessarily represent those of the US Treasury or the US government.

1 Introduction

The Federal Reserve (Fed) conducts monetary policy to achieve maximum employment and price stability. Within the Fed, the Federal Open Market Committee (FOMC) communicates decisions through post-meeting statements. Traditionally, the FOMC used open market operations to influence the federal funds rate. However, the 2008–2009 Great Recession necessitated an expansion into unconventional policy tools. Strategic statements were central to this transition. These statements manage market expectations of future policy and economic conditions. Wording changes in these announcements clearly shift expectations. However, the exact drivers of these movements remain unclear. Because post-meeting announcements are subject to an inherent communication bandwidth constraint, the Committee faces a zero-sum trade-off in statement real estate. It remains unexamined whether financial markets are responding to changes in explicit guidance on monetary policy tools, the shifting density of macroeconomic news, or the structural reallocation between the two.

I find that the reallocation of discussion intensity within FOMC statements between economic conditions and monetary policy tools has distinct effects on market expectations. To quantify these compositional shifts, I measure variations in "communication bandwidth" across sequential announcements using large-language-model (LLM) probabilistic topic classification. This methodology captures the shifting proportions of language devoted to specific themes from one meeting to the next, subject to the statement's total length. This approach allows me to assess the relationship among relative narrative bandwidth, semantic novelty, and high-frequency changes in financial market expectations of the future federal funds rate.

My paper presents four key findings. First, I document a structural "see-saw" effect in narrative composition: a relative expansion in the topic share regarding monetary policy tools, which mechanically decreases the space allocated to discussing macroeconomic conditions, is associated with an increase in the magnitude of monetary policy surprises. Symmetrically, shifting the communication mix toward economic data context dampens these surprises. Because these shares operate on a zero-sum basis by construction, this finding establishes a joint compositional channel. Changes in market expectations can be driven by two complementary behavioural mechanisms: an increase in policy tool prominence may introduce a signalling vacuum regarding the Fed's reaction function, while the concurrent deletion of macroeconomic detail removes the informational context and clarity necessary for financial markets to cleanly anticipate the FOMC's actions.

Second, I find that this narrative see-saw in topic shares influences the variation in market expectations independently and on top of the overall semantic dissimilarity between sequential statements. To isolate these channels, I include interaction terms between topic share and semantic novelty. I orthogonalise these terms via a residual centring approach. When a statement is structurally rewritten, shifting the narrative mix toward economic conditions yields a mitigating "clarification effect", dampening the uncertainty typically injected by novel phrasing. Conversely, the positive interaction for the monetary policy tools index suggests an "amplification effect". Here, combining novel wording with an intense relative focus on policy instruments signals a more complex regime and heightens financial market sensitivity. Overall, these interactions demonstrate that the pricing power of linguistic bandwidth is highly conditional upon semantic novelty.

Third, the thematic signals remain statistically significant after controlling for the absolute change in the target federal funds rate. This demonstrates that narrative composition conveys information distinct from both literal word changes and the primary policy action. Incorporating this compositional trade-off provides substantial incremental value. A model combining relative communication bandwidth, semantic novelty, and the target rate change explains 33% of the variance in monetary policy surprises. This represents a 30% relative improvement in explanatory power over the policy action alone.

Fourth, I document the longitudinal evolution of FOMC statement content. Using probabilistic topic classification, I show that communication bandwidth shifted strategically over the past two decades. Discussions about macroeconomic conditions such as inflation and output remain prevalent throughout the sample period. However, a structural break occurred following the 2008–2009 Great Recession. Pre-crisis statements focused on the federal funds rate and conventional policy tools. Post-crisis communication then pivoted towards financial conditions and unconventional monetary policy discussion. This thematic focus has remained the majority of the FOMC's descriptive language for nearly a decade. More recently, the relative allocation between conventional policy tools and economic conditions has become highly consistent. This lack of variation suggests a standardised narrative structure in contemporary FOMC communication. These results reflect a long-term shift towards data-dependent forward guidance.

This paper contributes to the literature on the "information effect". This field investigates whether market reactions to central bank announcements reflect the revelation of private information regarding the macroeconomy. A central challenge remains disentangling "pure" monetary policy shocks from these information signals (e.g., Campbell et al., 2012; Nakamura and Steinsson, 2018; Jarociński and Karadi, 2020; and others). Recently, papers such as Bauer and Swanson (2023) challenge this channel. They argue that such movements primarily reflect the Fed's response to public news. I provide a novel textual approach to this debate by documenting shifts in "communication bandwidth" via the relative composition trade-off between macroeconomic context and policy instruments. My findings suggest that shifting the narrative mix toward policy tools at the expense of economic context amplifies market expectations about future monetary policy. This provides a direct, non-structural test of how thematic composition influences the clarity of policy announcements.

This paper joins a growing literature using text analysis to study central bank communication. These studies examine the effects of communication on expectations and the macroeconomy. Traditional methods often relied on word counts or sentiment dictionaries to classify text (e.g., Gentzkow, Kelly, and Taddy, 2019; Husted, Rogers, and Sun, 2020; Aruoba and Drechsel, 2024; Acosta, 2023; and others). These methods can miss the nuanced context and thematic structure of policy announcements. In contrast, modern neural networks capture the full context and interdependencies of language. Recent studies leverage these tools to refine measures of monetary policy shocks (e.g., Handlan, 2022b; Doh, Song, and Yang, 2023; Bianchi, Ludvigson, and Ma, 2024; Piller, Schranz, and Schwaller, 2025; and others). I extend this research by focusing on the allocation of "communication bandwidth". Prior studies often rely on categorical topic labelling with equal weighting (e.g., Fischer et al., 2023; Dunn et al., 2024). Instead, I employ probabilistic classification via a state-of-the-art LLM to assign unequal thematic weights at the sentence level. This approach accurately documents the FOMC's relative compositional trade-off in attention between topics. I provide a distinct measurement of how this narrative see-saw influences expectations independently of semantic changes in statement wording.

The remainder of this paper is organised as follows. Section 2 describes the data sources used to study the relationship between the topic composition of FOMC statements and expectations about future monetary policy. Section 3 details the natural language processing methodology used to quantify communication bandwidth and the text analysis technique used to calculate the semantic similarity between sequential FOMC statements. Section 4 documents the longitudinal evolution of FOMC statement content. Section 5 presents my regression specifications and empirical results. Finally, Section 6 concludes.

2 Data

The sample period for my analysis runs from May 1999 to October 2019. The start date coincides with the FOMC's adoption of the practice of issuing a statement after every scheduled meeting, regardless of whether policy rates were changed. The end date provides a pre-COVID-19 benchmark for the analysis. This choice deliberately avoids the structural breaks and unprecedented market dynamics associated with the subsequent global pandemic.[1] The following subsections detail the data used in this analysis. I provide descriptive information on the FOMC statements, financial market expectations about future monetary policy, and

[1]Future versions of this paper could extend the sample period by adapting the systematic estimation to incorporate FOMC statements released after October 2019.

the specific control variable utilised.

2.1 FOMC Statements

FOMC decisions are announced in a press release after its eight scheduled annual meetings. These post-meeting statements often serve as the initial and primary source of monetary policy news for market participants.[2] The FOMC occasionally holds unscheduled meetings, but I drop these statements from my sample. This exclusion is to ensure that measured asset price changes are driven by the statement's content, not by the surprise of an unscheduled meeting. I source all scheduled statements from the Board of Governors of the Federal Reserve System website, resulting in a sample size of 165 statements.[3]

Text Pre-processing To prepare the FOMC statements for the LLM, I pre-process all 165 statements by standardising them to plain UTF-8 text with uniform spacing. I also remove URLs, release timestamps, and the list of regional bank request approvals. Most importantly, I remove the FOMC member voting record from the end of each statement. Although voting records are known to affect stock markets (C Madeira and J Madeira, 2019), they have less impact on interest-rate assets. Given my paper constructs monetary policy surprises using interest-rate futures prices, I omit these records to focus on the thematic and semantic similarity of the FOMC's deliberative language.[4]

Content and Evolution of Statements The FOMC suggests its announcements are crafted to shape market expectations regarding future monetary policy and macroeconomic conditions (Board of Governors of the Federal Reserve System, 2021). Consequently, the Committee appears to deliberately select its language and the relative emphasis placed on various topics. A typical FOMC statement discusses current macroeconomic conditions, communicates the Committee's economic expectations, and concludes with the new Federal

[2] Appendix Figure C1 shows that U.S. interest spikes on Google Trends for phrases such as "FOMC meeting" and "FOMC statement" during scheduled meeting dates. Further support for financial market reactions coming primarily from the FOMC statements is the fact that the top search query results on Google Search direct to the Board of Governors of the Federal Reserve System website.

[3] https://www.federalreserve.gov/monetarypolicy/fomc.htm

[4] Future versions of this paper will explore including the voting records as inputs.

Funds and discount rates. Following the Great Recession of 2008–2009, these statements began to include discussions of unconventional monetary policy, such as quantitative easing. This post-Crisis period saw an increase in statement complexity and length, particularly as the Committee relied more on communication to influence market expectations when it could no longer use rate changes. As seen in Figure 1, statement length grew rapidly until 2014. This trend partially reverses after 2014, but statements remain longer on average than before the Great Financial Crisis.

The following example is an excerpt of the FOMC statement released on 01 August, 2018:

> Information received since the Federal Open Market Committee met in June indicates that the labor market has continued to strengthen and that economic activity has been rising at a strong rate. [...]
>
> The Committee expects that further gradual increases in the target range for the federal funds rate will be consistent with sustained expansion of economic activity, strong labor market conditions, and inflation near the Committee's symmetric 2 percent objective over the medium term. Risks to the economic outlook appear roughly balanced.
>
> In view of realized and expected labor market conditions and inflation, the Committee decided to maintain the target range for the federal funds rate at 1-3/4 to 2 percent. The stance of monetary policy remains accommodative, thereby supporting strong labor market conditions and a sustained return to 2 percent inflation.

2.2 Interest-rate Futures and Monetary Policy Surprises

I use intraday (tick) data for interest-rate futures, sourced from the Thomson Reuters Tick History database (LSEG). This high-frequency data is necessary to analyse the short event windows (e.g., 30 minutes) common in the high-frequency identification literature. Within these windows, the variation in interest-rate futures prices measures the change in expected interest rate paths in response to FOMC announcements, often called monetary policy surprises.

I select a range of futures contracts commonly used to construct high-frequency monetary policy surprises (Kuttner, 2001; Gürkaynak, Sack, and Swanson, 2005; Gertler and Karadi, 2015; Nakamura and Steinsson, 2018; Jarociński and Karadi, 2020; Handlan, 2022b; and others). Specifically, I use Federal Funds futures (FFF), which are liquid up to three months out ($FF1$–$FF4$) and allow for measuring expectations for the current and next FOMC meetings. I construct these monetary policy surprises within an event window 10 minutes before and 40 minutes after the release of an FOMC statement. 50 minutes is the median of the event window lengths found to capture the full reaction of financial markets to monetary policy announcements by Tran (2026).

The settlement price of an FFF contract is based on the average effective federal funds rate over its expiry month. The contract pays out 100 minus this average rate. Therefore, the market's expected average federal funds rate is implied by its price. Let $p_{\tau,t}^{FFj}$ be the price of the $(j-1)$ month-ahead FFF at time t for FOMC meeting τ. The expected average federal funds rate is then calculated as $100-p_{\tau,t}^{FFj}$. In order to transform the variation in FFF prices into changes in expectations over federal funds rates, I extend the work of Nakamura and Steinsson (2018).

I calculate the "current meeting" monetary policy surprise ($mp1$) for each FOMC announcement τ at time $t+40$ (i.e., 40 minutes afterwards) as:

$$mp1_{\tau,t+40} = \frac{m}{m-d}\left(FF1_{\tau,t+40} - FF1_{\tau,t-10}\right), \tag{1}$$

where $FF1$ is the implied federal funds rate from the current-month contract $(100-p_{\tau,t}^{FF1})$, d is the day of the announcement, and m is the number of days in the month. A special case applies if the FOMC announcement occurs in the last seven days of the month $(m-d+1 \leq 7)$. To avoid distortion from the large $\frac{m}{m-d}$ scaling factor, the surprise is calculated as the simple change in the next month's futures contract:

$$mp1_{\tau,t+40} = FF2_{\tau,t+40} - FF2_{\tau,t-10}, \text{if } m-d+1 \leq 7. \tag{2}$$

The surprise for the next scheduled FOMC meeting ($mp2$) is more complex, as it must isolate the new information for that specific meeting. First, the relevant futures contract (j) is identified, which corresponds to the month of the next scheduled meeting ($j \in \{2,3,4\}$).

The surprise is then calculated as:

$$mp2_{\tau,t+40} = \frac{m_2}{m_2 - d_2}\left\{[FFj_{\tau,t+40} - FFj_{\tau,t-10}] - \frac{d2}{m2}mp1_{\tau,t+40}\right\}, \tag{3}$$

where FFj is the implied rate from the relevant contract, d_2 and m_2 are the day of and total days in the month of the next meeting, and $mp1_{\tau,t+n}$ is the current-meeting surprise.

As with the current-meeting surprise, a special case applies if the next meeting falls in the last seven days of its month ($m_2 - d_2 + 1 \leq 7$). To avoid distortion, the surprise is calculated as the simple change in the following month's contract:

$$mp2_{\tau,t+40} = FF(j+1)_{\tau,t+40} - FF(j+1)_{\tau,t-10}, \text{if } m - d + 1 \leq 7. \tag{4}$$

Because I am investigating how the narrative see-saw of monetary policy announcements affects market expectations about future monetary policy, I focus on the absolute change of the current- and next-meeting surprises, $|mp1_{\tau,t+40}|$ and $|mp2_{\tau,t+40}|$, respectively. Descriptive statistics for both surprises can be found in Table 1.

A potential concern regarding the identification of monetary policy surprises is their predictability based on preceding macroeconomic news. Alam (2022) suggests that conventional surprises can be predicted by data releases within a five-day window prior to the FOMC meeting. Furthermore, Handlan (2022a) documents that a minority of sample dates coincide with morning Consumer Price Index (CPI) releases. However, these factors are unlikely to invalidate my analysis of relative communication bandwidth for two reasons.

First, while the magnitude of a surprise may exhibit some predictability, this paper focuses on the thematic composition of the statement itself. Even if a policy shift is anticipated, the FOMC's specific linguistic choice to bias emphasis toward either macroeconomic indicators or policy instruments provides a contemporaneous channel of information. This channel directly influences how that surprise is formed. Second, my empirical strategy relies on a narrow 50-minute window centred on the statement release. In an efficient market, any predictability from prior data should already be incorporated into pre-statement futures prices. Therefore, the variation captured within this high-frequency window remains a direct response to the relative compositional trade-off between the topics of the FOMC announcement, rather than a delayed reaction to earlier macroeconomic news.

2.3 Control Variable: Target Federal Funds Rate

A standard approach in the monetary policy literature is to regress changes in expectations on changes in the target federal funds rate. This specification assumes the target rate change is a sufficient statistic for the informational content of the FOMC statement. However, statements also encompass discussions of current economic conditions and forward guidance. These components are not necessarily reflected in the rate decision.[5] A more robust rationale for including the target rate is its simultaneous announcement with the statement. In other words, the target rate changes within the same event window and therefore serves as a necessary control in event-study analysis.

These data are sourced from Federal Reserve Economic Data (FRED). Since December 2008, the FOMC has reported a target range for the federal funds rate rather than a single point. Because this range is constant in width and varies only by level, I utilise the midpoint of the reported range to extend the target rate series through October 2019. Table 1 presents descriptive statistics for the change in the target federal funds rate (Δr) and its absolute value ($|\Delta r|$).

3 The Topic and Semantic Content of FOMC Statements

Quantifying the thematic and semantic content of FOMC statements requires a precise measurement of how the Committee allocates its limited communication bandwidth. Constructing the textual signals needed for this analysis demands a methodology capable of capturing

[5] I provide further evidence against this assumption in Section 4.

the full information content of each announcement as a relative compositional trade-off. In other words, the methodology must formalise the multidimensional nature of policy communication, mapping how the balance shifts from assessments of current macroeconomic conditions to signals regarding future policy instruments.

Traditional text analysis methods, such as "fitting predictive models on simple counts of text features" (Gentzkow, Kelly, and Taddy, 2019), typically fail to capture the full information content of text. These methods often miss context and word interdependencies. For example, a "bag-of-words" approach would treat the phrases "*employment* went up, but *inflation* did not" and "*inflation* went up, but *employment* did not" as identical. Similarly, "n-gram" methods, which measure neighbouring words, miss information spread across a sentence. A trigram might capture "economic growth slowed" but miss the crucial reversal in the phrase "economic growth slowed, but is expected to pick up pace later this year". Therefore, these popular methods cannot realistically capture the "full" information content of FOMC statements.[6]

In contrast, text-analysis neural networks such as LLMs *can* capture complex relationships like context and interdependencies. This capability is essential for accurately identifying the thematic focus of FOMC statements, where the meaning of a sentence often depends on the relationship between non-adjacent terms, such as "economic growth" and "pick up pace". The power of these networks is mathematically justified by the Universal Approximation Theorem (Hornik, Stinchcombe, and White, 1989; and others), which states that any continuous function (such as the classification of text into thematic topics) can be approximated by a network with at least one hidden layer to any desired accuracy.[7] In modern architectures, self-attention mechanisms perform this universal approximation capacity by processing full linguistic sequences simultaneously. Given their success in other social sci-

[6]Papers like Babur and Cleophas (2017) have shown that model performance does not increase monotonically with the size of "n-gram" methods.

[7]The Universal Approximation Theorem is an existence theorem; it guarantees *existence* of a network with at least one hidden layer, but does not specify its structure. In practice, deep networks (multiple layers) are often used to reduce parameters and computational requirements for complex approximations.

ences, these methods offer similar benefits for studying monetary policy communication in economics (Gentzkow, Kelly, and Taddy, 2019).

3.1 Gemini 3.1 Pro, a State-of-the-art LLM for Text Analysis

I employ Gemini 3.1 Pro (Google DeepMind and Google AI, 2026), a state-of-the-art large language model, for the sentence-level thematic classification of the FOMC corpus. The model features a natively multimodal architecture with a dynamic "three-tier thinking" system. This design allows it to modulate computational depth based on reasoning complexity.[8] Unlike earlier encoder models such as XLNet, Gemini 3.1 Pro utilises a "Mixture-of-Experts (MoE)" structure optimised for deep semantic synthesis.[9] This structure makes the model uniquely suited to identifying the subtle, non-adjacent linguistic cues in FOMC communication. Rather than task-specific fine-tuning, I leverage its native reasoning to extract probabilistic topic shares $p \in [0, 1]$ for each sentence. This approach ensures measurements reflect the full density of the Committee's thematic focus. Importantly, these probabilistic shares capture a more granular representation of communication bandwidth than binary classification methods can provide.

3.2 Probabilistic Topic Classification by Google Gemini 3.1 Pro

To extract the thematic content of the FOMC statements, I develop a structured prompting architecture that leverages the long-context capabilities of Gemini 3.1 Pro. Rather than processing sentences in isolation, I provide the model with the entire statement as context before classifying its constituent sentences. This "whole-document context" allows the model to resolve ambiguous references. It ensures thematic labels are consistent with the surrounding narrative intent. As a result, the model interprets the statement similarly to a human

[8] The Gemini 3 series supports an input window of up to one million tokens and an output ceiling of 64,000 tokens. See Google AI for Developers for architectural details.

[9] LLMs like Gemini 3.1 Pro leverage "in-context learning" to perform sophisticated classification with minimal supervision. They maintain superior performance on abstract logic benchmarks like ARC-AGI-2.

reader. My sample of FOMC statements yields 2,054 sentences.

The classification procedure is governed by system instructions that assign the model the persona of an expert macroeconomist participating in an experiment. The model is provided with a taxonomy and non-exhaustive term lists for six mutually exclusive categories following Dunn et al. (2024): Real Activity, Labour Market, Inflation, Financial Conditions, Fed Funds Rate, and Balance Sheet. These instructions dictate the core logic Gemini 3.1 Pro must follow. For example, the neural network must disregard external information to minimise look-ahead bias and utilise the English grammatical structure to perform syntactic weighting.[10] For each sentence, the model performs a probabilistic assignment $p \in [0, 1]$ across these categories, constrained to sum to one. This probability distribution represents the model's assessment of how a human expert would allocate the thematic weight of the sentence.

To illustrate this probabilistic assignment, I provide two sentence examples from the corpus. First, a sentence with a singular topic focus receives a categorical assignment. Sentence 5 from the 5 October, 2010 FOMC statement is assigned a probability of 1.0 for Inflation:

> Moreover, the increase in energy prices, though having limited effect on core measures of prices to date, poses a risk of raising inflation expectations.

In this instance, because "energy prices" and "core measures" are mentioned, the model correctly identifies that the entire sentence is directed toward the singular latent variable of inflation risk.

Conversely, sentences often convey a multidimensional policy signal, which require an unequal distribution of topic shares. Consider Sentence 4 from the 22 August, 2000 FOMC statement:

> Nonetheless, the Committee remains concerned about the risk of a continuing gap between the growth of demand and potential supply at a time when the utilisation of the pool of available workers remains at an unusually high level.

[10] The full text of the system instructions and structured prompt template are provided in Appendix A.

For this sentence, the network assigns a probability of 0.6 to Real Activity and 0.4 to the Labour Market topic. This assignment reflects the model's ability to perform syntactic weighting. The primary subject and object (i.e., the gap between demand and supply) represent Real Activity. The temporal clause regarding the "pool of available workers" is identified as a secondary Labour Market signal, which provides the necessary context for the Committee's primary concern.

To ensure replicability, I set the temperature to zero and utilise a fixed seed of 47. These parameter values force the neural network to follow a deterministic decoding path. The model consistently selects the highest-probability output at each step. This process completely eliminates stochastic variation across runs. Consequently, this approach ensures the classification remains strictly based on the model's most certain interpretation.

3.3 Model Validation Against Human Ground Truth

To assess the reliability of the probabilistic topic classification, I perform a validation exercise. I compare the topic labels generated by Gemini 3.1 Pro against a manually coded "ground truth" dataset created in two steps. First, I randomly sample 10% of the sentences for manual assignment. Second, I perform a double-blind assignment. I independently code each sentence without knowledge of the model's output. The neural network remains strictly uninformed of my manual labels. To ensure consistency in the human assignment, I utilise a formal coding guidebook based on macroeconomic theory and English syntactic structure. This allows for nuanced, non-binary topic shares with unequal weighting (e.g., 1.0, 0.5/0.5, or 0.7/0.3). Such shares capture instances where a sentence conveys multiple policy signals.[11]

I evaluate the model's performance by comparing its zero-shot classifications against my expert assignments across the six topics. As shown in Table 2, the results demonstrate an exceptional alignment with human intuition. The average and median Pearson correlations across all topics are 0.9311 and 0.9434, respectively. Furthermore, the network achieves a low

[11] The complete guidebook and the specific coding rules used for the human validation are provided in Appendix B.

average and median mean squared error (MSE) of 0.0120 and 0.0082, respectively. These metrics indicate that Gemini 3.1 Pro effectively replicates the "ground truth" labels of a trained macroeconomist without task-specific fine-tuning. This strong correlation suggests that the probabilistic outputs provide a robust and objective foundation for analysing FOMC communication bandwidth.

3.4 Aggregation and Measuring Communication Bandwidth

I aggregate the probabilistic topic shares across each FOMC statement to construct meeting-level measures of communication. This paper assumes that the "communication bandwidth" dedicated to a topic is proportional to the relative volume of language the FOMC allocates to it. Accordingly, I weight each sentence by its word count. This weighting function map the text into a finite, zero-sum compositional space. This approach ensures that detailed thematic shifts carry greater weight in the aggregate measures.

Formally, for a given FOMC statement released at time τ consisting of n sentences, let $p_{i,k,\tau} \in [0,1]$ represent the probability assigned by Gemini 3.1 Pro to topic k for sentence i. Let $w_{i,\tau}$ be the word count of that sentence. The statement-level topic share (i.e., communication bandwidth) for topic k is the word-weighted average:

$$W_{k,\tau} = \frac{\sum_{i=1}^{n} (p_{i,k,\tau} w_{i,\tau})}{\sum_{i=1}^{n} w_{i,\tau}}, \tag{5}$$

where $\sum_{k=1}^{6} W_{k,\tau} = 1$. This aggregation results in a six-dimensional vector $\mathbf{W}_\tau$ representing the thematic density of the announcement. By adjusting for word-count intensity, this measure captures both the presence of a topic and the depth of the Committee's descriptive focus. This construction allows for an objective tracking of how the FOMC reallocates its limited communication bandwidth within the statement. For example, the measure captures the zero-sum shift from interest-rate guidance to balance-sheet policy following the 2008–2009 Great Recession.

I compute the first difference of the statement-level topic shares, $\Delta W_{k,\tau} = W_{k,\tau} - W_{k,\tau-1}$, to examine the FOMC's communication strategy. This measure captures the inter-meeting reallocation of bandwidth. Specifically, it reveals how the FOMC shifts its descriptive focus between topics over time. While the levels of $W_{k,\tau}$ provide a snapshot of the Committee's language priorities, these first differences highlight marginal adjustments in topic intensity. This specification cleanly captures compositional shifts, such as a direct reallocation from discussions about the Labour Market to Inflation.

3.5 The Semantic Content of FOMC statements

FOMC statements, while relatively concise, have language and semantics that vary over time (Acosta and Meade, 2015; Handlan, 2022a). I measure similarity using a standard bag-of-words model, which represents each statement as a vector of weighted word frequencies. This method assumes that documents with similar word frequencies are discussing similar topics, allowing me to calculate a similarity score between any two statements.

The specific weighted frequency I use is Term Frequency-Inverse Document Frequency (TFIDF), implemented via the Python `sklearn` library.[12] TFIDF computes a weight for each word by balancing its frequency within a single document (Term Frequency) against its frequency across all documents (Inverse Document Frequency).

This weighting serves a crucial purpose: common words that provide little unique meaning (e.g., "a", "the") appear in all documents, so their high document-wide frequency significantly reduces their final weight.

Similarly, standard policy words (e.g., "unemployment", "inflation") also have their weights reduced, though less so, as their relative frequency can still differentiate statements. Conversely, unique or rare words (e.g., "persists", "tight") that appear in few documents receive a higher weight, as they are more informative. These differential weights are what allow the vector analysis to effectively measure the degree of similarity between any two FOMC

[12] The `sklearn` library utilises specific smoothing and normalisation constants that differ slightly from the standard "textbook" definitions. For clarity, I document the generalised textbook formulas in this section.

statements.

The calculation of the TFIDF matrix is as follows: Let D be the set of FOMC statements and T be the set of all terms. Let a single document and term be indexed by $d \in D$ and $t \in T$, respectively. The term frequency is:

$$tf_{d,t} = \ln\left(\frac{tc_{d,t}}{nt_d}\right) + 1, \tag{6}$$

where $tc_{t,d}$ is the count of term t in document d, nt_d is the total number of words in d, and the log and addition terms smooth the frequency measure. The inverse document frequency is:

$$idf_{d,t} = \ln\left(\frac{nd}{df_{d,t} + 1}\right) + 1, \tag{7}$$

where nd is the total number of documents and $df_{t,d}$ is the number of documents in which term t appears. Multiplying the two terms gives the final weighted frequency:

$$TFIDF_{d,t} = tf_{d,t} \times idf_{d,t}. \tag{8}$$

Calculating this value for all terms and documents yields a $D \times T$ matrix, where each element $TFIDF_{d,t}$ is the final weighted frequency. Essentially, the higher the TFIDF value for a given term, the more informative that term is for distinguishing the content of a specific FOMC statement from all others.

The TFIDF matrix from the *current* pre-processed text would be inaccurate. This simple model is not sophisticated enough to handle context, meaning it would incorrectly treat "Federal Funds Rate" and "federal funds rate" as different terms. I mitigate this with further pre-processing. First, I convert all text to lowercase. Second, I remove common "stop words" (articles, pronouns, conjunctions) that convey little semantic meaning.[13] Third, I convert all words to their "base" or "stem" form (e.g., "increases", "increasing", and "increase" all

[13]I retain numbers, as they are crucial for conveying semantic content like interest rates or forward guidance.

become "increas"). This last step allows for the document similarity matrix to provide a more meaningful interpretation of how semantically persistent FOMC statements are over time.

Calculating the TFIDF matrix on this fully cleaned sample (May 1999–October 2019) yields a final matrix of 165 rows (statements) and 966 columns (terms).

With the TFIDF matrix, I can now measure the similarity between any two FOMC statements. I create a document similarity matrix by multiplying the TFIDF matrix by its transpose:

$$\text{Document Similarity Matrix} = TFIDF \cdot TFIDF^T. \tag{9}$$

Because the TFIDF vectors are normalised, this multiplication is equivalent to calculating the dot product between every pair of document row vectors. This allows me to define the similarity between any two statements (A and B) as their *cosine similarity*:

$$S^{A,B} := \cos\theta = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\|\|\mathbf{B}\|}, \tag{10}$$

where $\mathbf{A}$ and $\mathbf{B}$ are the TFIDF vectors for statements A and B. A cosine similarity of one means the statements are identical, while a value of zero means they have no common informative terms. The more informative base terms two statements share, the closer their cosine similarity will be to one.

The Document Similarity Matrix allows me to compare any two statements. I focus on the similarity between sequential statements, $(d, d-1)$, which I denote S^1 and plot in Figure 2.[14] Figure 2 shows that the cosine similarity between sequential FOMC statements increases over time. This trend of increased standardisation in FOMC language begins around 2009–2010, coinciding with the federal funds rate approaching the zero lower bound. As discussed in Handlan (2022a), one explanation is that the FOMC sought to reduce market surprises by standardising its statement vocabulary during and after the financial crisis. This trend

[14] Appendix Figure C2 presents the full Document Similarity Matrix as a heat map.

of high similarity continued through 2019, even after the zero lower bound ended, reflecting the FOMC's broader efforts to increase policy transparency under Chairs Yellen and Powell.

For regression analysis, I transform the semantic similarity measure between sequential FOMC statements into measuring document dissimilarity, D^1:

$$D^1 = 1 - S^1. \tag{11}$$

Descriptive statistics for both S^1 and D^1 can be found in Table 1.

4 The Evolution of FOMC Communication Bandwidth

This section tracks the evolution of the FOMC's thematic focus from May 1999 to October 2019. By aggregating sentence-level probabilities into statement-level shares, I quantify how the FOMC reallocates its limited communication bandwidth. By construction, these shares sum to one for every statement, reflecting a zero-sum narrative trade-off. Figure 3 illustrates these statement-level shares over time. Vertical dashed lines depict the respective tenures of Fed Chairs Greenspan, Bernanke, Yellen, and Powell.

The visual evidence in Figure 3 reveals distinct shifts in narrative priorities across leadership regimes. During the final Greenspan years and early Bernanke years, the FOMC's bandwidth was primarily occupied by a dual focus on Real Activity and Inflation. A structural pivot in the narrative occurred at the onset of the 2008–2009 Great Recession. Specifically, in late 2008, the Balance Sheet and Financial Conditions topics began to occupy a significant portion of the communication bandwidth. These topics were previously negligible, and their rise signifies the historical transition to unconventional monetary policy. Interestingly, the distribution of bandwidth across Real Activity, the Labour Market, Inflation, and the Fed Funds Rate appears notably even during the Yellen and early Powell years. This balanced allocation aligns with the "semantic standardisation" documented by Handlan (2022a). By maintaining a steady focus across the core pillars of the dual mandate, the Committee effectively institutionalised a highly predictable narrative structure during the post-crisis era.

The informational content of these statement-level topic shares is empirically distinct from both semantic dissimilarity and changes in the target federal funds rate. As shown by the Pearson correlations in Table 3, the FOMC's communication bandwidth conveys information that is not fully captured by literal wording variation or interest rate adjustments. Consistent with the findings in Handlan (2022a), these results indicate that the Committee's descriptive narrative serves as a distinct policy instrument.

For the following regression analyses, the six individual topic shares are aggregated into two composite thematic indices. This step collapses the communication simplex into a direct structural trade-off between economic context and policy instruments. The economic context index (EC) sums the shares of Real Activity, Labour Market, Inflation, and Financial Conditions. The monetary policy index (MP) sums the shares of the Fed Funds Rate and the Balance Sheet. This restricted framework models the narrative see-saw, examining how the reallocation of communication bandwidth shapes the variation in market expectations about future monetary policy. Descriptive statistics for all six thematic categories and the two composite indices are presented in Table 1.

5 Bandwidth and the Magnitude of MP Expectations

In this section, I evaluate how FOMC communication bandwidth impacts the scale of monetary policy surprises. I first outline the identification strategy used to address potential endogeneity and simultaneity challenges. I then detail the baseline regression specifications. Finally, I present the empirical results and discuss the underlying economic mechanisms.

5.1 Identification Strategy

This analysis addresses endogeneity and simultaneity concerns through high-frequency identification. The Federal Reserve naturally adjusts its topic intensities based on the prevailing economic environment. However, measuring asset price surprises within a narrow 50-minute window isolates the unanticipated component of the announcement. Asset prices ten minutes before the release incorporate all existing public information. Consequently, any asset market movement within this narrow window is attributed solely to new information contained in the FOMC announcement.

Simultaneity is mitigated by the temporal sequence of the release. The FOMC finalises the statement text prior to publication. Because the market surprise materialises only after the text becomes public, the surprise magnitude cannot influence the thematic intensities already committed to the document. Following Handlan (2022a), I control for the change in the target federal funds rate. This approach isolates the expectation shift driven specifically by the narrative see-saw. As a result, the estimates reflect narrative reallocation rather than simultaneous adjustments to the policy rate.

5.2 Empirical Results

I use three regression specifications to evaluate whether FOMC communication bandwidth affects the magnitude of market expectations about future monetary policy. Specifically, I estimate ordinary least squares regressions for current-meeting and next-meeting monetary policy surprises. The core independent variable is the first difference of the thematic index for monetary policy tools. Because the two thematic indices form a communication simplex, I omit the economic conditions index to avoid perfect multicollinearity. Focusing on changes in communication weights rather than levels isolates how a thematic pivot affects expectation variation. This design ensures the coefficients capture the high-frequency market reaction to a structural reallocation of informational priorities. All specifications have the right-hand side variables mean-centred to address potential non-essential multicollinearity. Furthermore, all specifications employ heteroscedasticity and autocorrelation robust standard errors (Newey and West, 1987; Piazzesi and Swanson, 2008).

The first specification regresses the absolute change in current- and next-meeting monetary policy surprises on the first difference of the monetary policy tools index:

$$|mpj_{\tau,t+40}| = \beta_0 + \beta_1 \Delta MP_\tau + \varepsilon_\tau, \tag{12}$$

where $j \in \{1, 2\}$ denotes the current-meeting or next-meeting surprise, both surprises are measured within an event window starting 10 minutes before and ending 40 minutes after a statement release, and τ represents the FOMC meeting date. This baseline regression identifies the continuous marginal impact of a relative narrative shift toward policy instruments. Because the underlying shares are bound by the simplex, β_1 directly captures the effect of tilting the narrative see-saw away from economic context.

The second specification examines the interaction between thematic indices and semantic dissimilarity, D^1_τ. I implement a residual centring procedure following Little, Bovaird, and Widaman (2006). First, the raw product of the monetary policy index change and the continuous semantic dissimilarity metric D^1_τ is regressed on its constituent linear components:

$$(\Delta MP_\tau \cdot D^1_\tau) = \alpha_0 + \alpha_1 \Delta MP_\tau + \alpha_2 D^1_\tau + \tilde{I}_\tau, \tag{13}$$

where $\tilde{I}_\tau$ represents the orthogonalised interaction residual. This residual is subsequently included in the main regression to isolate the marginal effect of shifts in thematic focus:

$$|mpj_{\tau,t+40}| = \gamma_0 + \gamma_1 \Delta MP_\tau + \gamma_2 D^1_\tau + \gamma_3 \tilde{I}_\tau + \varepsilon_\tau, \tag{14}$$

where $j \in \{1, 2\}$ denotes the current-meeting or next-meeting surprises, both surprises are measured within an event window starting 10 minutes before and ending 40 minutes after a statement release, and τ represents the FOMC meeting date. The coefficient γ_3 captures the marginal impact when a reallocation of communication bandwidth occurs alongside a significant structural rewrite of the statement text.

The final specification incorporates the absolute change in the target federal funds rate,

$|\Delta r_\tau|$, as a control variable. This model tests whether communication bandwidth remains associated with the variation in market expectations after accounting for literal wording changes and the direct policy rate signal. The model is specified as follows:

$$|mpj_{\tau,t+40}| = \delta_0 + \delta_1 \Delta MP_\tau + \delta_2 D^1_\tau + \delta_3 \tilde{I}_\tau + \delta_4 |\Delta r_\tau| + \varepsilon_\tau, \tag{15}$$

where $j \in \{1, 2\}$ denotes the current-meeting or next-meeting surprises, both surprises are measured within an event window starting 10 minutes before and ending 40 minutes after a statement release, and τ represents the FOMC meeting date. By including $|\Delta r_\tau|$, this specification investigates the extent to which thematic content influences market expectations independently of the primary monetary policy action, literal wording changes, and the interaction between thematic and semantic content.

Results for the first specification are presented in columns (1) and (4) of Table 4. They reveal a statistically significant relationship between communication bandwidth reallocation and the magnitude of market expectations. A thematic pivot toward monetary policy tools (i.e., ΔMP_τ) by one standard deviation is associated with a 0.28 and 0.25 standard-deviation increase in the magnitude of current- and next-meeting expectations, respectively.[15] This baseline result demonstrates that tilting the communication see-saw away from economic context and toward policy instruments directly heightens the scale of monetary policy surprises.

These results highlight a fundamental trade-off in the FOMC's communication strategy. Because thematic shares must sum to unity, any increased focus on policy instruments necessarily crowds out the descriptive bandwidth available for macroeconomic conditions. The findings suggest that providing more detail on the economic environment helps the market identify which macroeconomic indicators are currently salient to the Committee's decision-making process. Conversely, prioritising the discussion of policy tools over the underlying

[15] Specifically, a one-standard-deviation increase in ΔMP_τ increases $|mp1_{\tau,t+40}|$ by $\left[\beta_1 \frac{\mathrm{std}(\Delta MP_\tau)}{\mathrm{std}(|mp1_\tau|)}\right] \approx 0.28$ standard deviations.

"why" of the economic data may make it more difficult for market participants to align their expectations with the Committee's framework. Accordingly, these baseline results indicate that the thematic allocation of the FOMC statement, on its own, explains approximately 8% and 6% of the variance in current- and next-meeting monetary policy surprises, respectively.

Results for the second specification are reported in columns (2) and (5) of Table 4. This model examines whether the impact of communication bandwidth operates independently of literal wording changes. Even after controlling for semantic dissimilarity D^1_τ, the coefficient for the monetary policy index remains statistically significant at the 5% level. A one-standard-deviation increase in ΔMP_τ is associated with a 0.24 standard-deviation increase in the magnitude of current-meeting surprises. While D^1_τ is a strong predictor of asset price volatility, its inclusion does not negate the explanatory power of the thematic focus. Crucially, the orthogonalised interaction residual $\tilde{I}_\tau$ is statistically insignificant across both specifications. This suggests that the relative intensity of the topic influences expectations distinctly from the structural novelty of the statement's vocabulary. These findings reinforce the argument that "what" the FOMC discusses is as relevant as "how" the Committee phrases it.

Although the interaction residuals $\tilde{I}_\tau$ are not statistically significant, their positive signs provide directional evidence regarding the conditional effects of narrative reallocation. The positive point estimates suggest that a simultaneous shift toward policy instruments and novel phrasing may amplify market surprises. These directional results suggest that the market's interpretation of the Committee's thematic focus is contingent upon the linguistic delivery of the statement. However, these coefficients are entirely swamped by their standard errors. This lack of statistical power is a direct consequence of the normalised simplex constraint, which forces a rigid relative trade-off across text categories. This limitation strongly motivates the un-normalised specifications presented later in the paper. Breaking the zero-sum constraint allows the non-linear interaction residuals to capture absolute scale expansions cleanly during structural rewrites.

The marginal contribution of communication bandwidth is evident when comparing these results to baseline language models. Semantic dissimilarity alone explains approximately 16% of the variance in current-meeting surprises, as reported in Appendix Table D1. Incorporating the thematic index and the interaction residual increases the R^2 to 22%. This change represents an approximate 40% increase in explanatory power. This gain indicates that the thematic mix of the statement provides substantial information for market participants that is not captured by literal wording novelty alone.

The final specification adds the absolute change in the target federal funds rate, $|\Delta r_\tau|$, as a control variable. This model tests whether the thematic signal remains distinct from both the primary policy action and literal wording changes. As reported in columns (3) and (6) of Table 4, the coefficients for the monetary policy index remain statistically significant. A one-standard-deviation increase in ΔMP_τ is associated with a 0.18 standard-deviation increase in current-meeting surprises. For next-meeting expectations, this amplifying effect is 0.13 standard deviations. This survival demonstrates that the narrative see-saw conveys distinct policy information independently of the underlying target rate shock.

The incremental value of these textual measures is evident when compared to the baseline models. As documented in Appendix Table D1, the policy action $|\Delta r_\tau|$ alone explains approximately 25% of the variance in current-meeting surprises. Incorporating the thematic and semantic text measures increases the R^2 to 33%, representing an approximate 30% relative increase in explanatory power. Furthermore, this full model provides a 10% gain over a specification that combines the target rate shock with semantic dissimilarity.[16] These results indicate that accounting for thematic focus provides a more comprehensive understanding of how market participants update expectations. The narrative see-saw captures distinct information omitted by regressions relying solely on policy actions or literal wording novelty.

[16] While Handlan (2022a) focuses on the interaction between wording and policy actions, my objective is to isolate the independent contribution of communication bandwidth. I include D^1_τ and $|\Delta r_\tau|$ as linear controls to demonstrate that the allocation of discussion topics provides a distinct signal. This signal is not subsumed by wording novelty or the interest rate move itself.

6 Conclusion

This paper introduces a novel framework for quantifying the thematic focus of FOMC communication. By applying large-language-model probabilistic topic classification to FOMC statements, I construct indices that measure the allocation of communication bandwidth across economic and monetary policy themes. This approach provides a structural view of the Committee's narrative that literal wording novelty measures alone cannot fully capture.

The temporal analysis reveals that the FOMC's narrative priorities shift significantly across economic business cycles. A structural pivot occurred during the 2008 financial crisis, as communication bandwidth reallocated from macroeconomic indicators toward unconventional policy tools. In contrast, the post-crisis era exhibits a more balanced and consistent thematic distribution. These narrative see-saw dynamics are empirically distinct from both literal wording changes and interest rate adjustments. This evidence confirms that the Committee's descriptive narrative serves as an independent policy instrument.

The empirical results demonstrate that the thematic mix of the FOMC statement is a significant driver of market expectations. Because the narrative see-saw operates as a zero-sum allocation, its movements reveal a fundamental trade-off in the Committee's communication strategy. A relative pivot toward monetary policy tools increases the magnitude of high-frequency surprises. Conversely, a symmetric reallocation toward macroeconomic conditions dampens the scale of these expectations. Providing descriptive detail on the economic environment helps market participants identify salient macroeconomic indicators. This context assists the public in tracking the Committee's underlying reaction function. However, prioritising explicit policy tools without sufficient economic context removes this background and directly amplifies the magnitude of monetary policy surprises.

The narrative see-saw provides substantial incremental explanatory power regarding the magnitude of market expectations. Incorporating this thematic communication index increases the explained variation in current-meeting surprises by approximately 30% compared to models utilising the target rate shock alone. Furthermore, the thematic signal remains statistically significant across all models. This significance survives after controlling for literal wording changes and adjustments to the target rate. Ultimately, this paper demonstrates that the substance of the FOMC's thematic focus is as consequential as its choice of phrasing. Accounting for this thematic allocation provides a more comprehensive understanding of how central bank communication informs and shapes financial market expectations.

References

Acosta, Miguel (2023). "The Perceived Causes of Monetary Policy Surprises". Published manuscript.

Acosta, Miguel and Ellen E. Meade (2015). *Hanging on Every Word: Semantic Analysis of the FOMC's Postmeeting Statement*. Tech. rep. FEDS Notes. Board of Governors of the Federal Reserve System.

Alam, Zohair (2022). "Learning About Fed Policy From Macro Announcements: A Tale of Two FOMC Days". SSRN Working Paper No 4065084.

Aruoba, S. Borağan and Thomas Drechsel (2024). *Identifying Monetary Policy Shocks: A Natural Language Approach*. Working Paper 32417. National Bureau of Economic Research.

Babur, Önder and Loek Cleophas (2017). "Using n-grams for the Automated Clustering of Structural Models". In: *SOFSEM 2017: Theory and Practice of Computer Science*. Ed. by Bernhard Steffen, Christel Baier, Mark van den Brand, Johann Eder, Mike Hinchey, and Tiziana Margaria. Cham: Springer International Publishing, pp. 510–524.

Bauer, Michael D. and Eric T. Swanson (2023). "A Reassessment of Monetary Policy Surprises and High-Frequency Identification". In: *NBER Macroeconomics Annual* 37.1, pp. 87–155.

Bianchi, Francesco, Sydney C Ludvigson, and Sai Ma (2024). *What Hundreds of Economic News Events Say About Belief Overreaction in the Stock Market*. Working Paper 32301. National Bureau of Economic Research.

Board of Governors of the Federal Reserve System (2021). "The Fed Explained: What the Central Bank Does". In: *Public Education & Outreach*. 11th ed. Federal Reserve System Publication.

Campbell, Jeffrey R., Charles L. Evans, Jonas D. M. Fisher, and Alejandro Justiniano (2012). "Macroeconomic Effects of Federal Reserve Forward Guidance". In: *Brookings Papers on Economic Activity* 43.1 (Spring 2012), pp. 1–80.

Doh, Taeyoung, Dongho Song, and Shu-Kuei Yang (2023). "Deciphering Federal Reserve Communication via Text Analysis of Alternative FOMC Statements". Federal Reserve Bank of Kansa City Working Paper.

Dunn, Wendy, Ellen E. Meade, Nitish Sinha, and Raakin Kabir (2024). *Using Generative AI Models to Understand FOMC Monetary Policy Discussions*. Tech. rep. FEDS Notes. Board of Governors of the Federal Reserve System.

Fischer, Eric, Rebecca McCaughrin, Saketh Prazad, and Mark Vandergon (2023). *Fed Transparency and Policy Expectation Errors: A Text Analysis Approach*. Tech. rep. 1081, November. Staff Reports. Federal Reserve Bank of New York.

Gentzkow, Matthew, Bryan Kelly, and Matt Taddy (2019). “Text as Data”. In: *Journal of Economic Literature* 57.3, pp. 535–74.

Gertler, Mark and Peter Karadi (2015). “Monetary Policy Surprises, Credit Costs, and Economic Activity”. In: *American Economic Journal: Macroeconomics* 7.1, pp. 44–76.

Google DeepMind and Google AI (2026). *Gemini 3.1 Pro Model Card*.

Gürkaynak, Refet S., Brian Sack, and Eric T. Swanson (2005). “Do Actions Speak Louder Than Words? The Response of Asset Prices to Monetary Policy Actions and Statements”. In: *International Journal of Central Banking* 1.1.

Handlan, Amy (2022a). “FedSpeak Matters: Statement Similarity and Monetary Policy Expectations”. Published manuscript.

Handlan, Amy (2022b). “Text Shocks and Monetary Surprises: Text Analysis of FOMC Statements with Machine Learning”. Published manuscript.

Hornik, Kurt, Maxwell Stinchcombe, and Halbert White (1989). “Multilayer feedforward networks are universal approximators”. In: *Neural Networks* 2.5, pp. 359–366.

Husted, Lucas, John Rogers, and Bo Sun (2020). “Monetary policy uncertainty”. In: *Journal of Monetary Economics* 115, pp. 20–36.

Jarociński, Marek and Peter Karadi (2020). “Deconstructing Monetary Policy Surprises—The Role of Information Shocks”. In: *American Economic Journal: Macroeconomics* 12.2, pp. 1–43.

Kuttner, Kenneth N. (2001). “Monetary Policy Surprises and Interest Rates: Evidence from the Fed Funds Futures Market”. In: *Journal of Monetary Economics* 47.3, pp. 523–544.

Little, Todd D., James A. Bovaird, and Keith F. Widaman (2006). “On the Merits of Orthogonalizing Powered and Product Terms: Implications for Modeling Interactions Among Latent Variables”. In: *Structural Equation Modeling: A Multidisciplinary Journal* 13.4, pp. 497–519. eprint: https://doi.org/10.1207/s15328007sem1304_1,.

Madeira, Carlos and João Madeira (2019). “The Effect of FOMC Votes on Financial Markets”. In: *The Review of Economics and Statistics* 101.5, pp. 921–932.

Nakamura, Emi and Jón Steinsson (2018). “High-Frequency Identification of Monetary Non-Neutrality: The Information Effect”. In: *The Quarterly Journal of Economics* 133.3, pp. 1283–1330.

Newey, Whitney K. and Kenneth D. West (1987). “A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix”. In: *Econometrica* 55.3, pp. 703–708.

Piazzesi, Monika and Eric T. Swanson (2008). “Futures prices as risk-adjusted forecasts of monetary policy”. In: *Journal of Monetary Economics* 55.4, pp. 677–691.

Piller, Alexander, Marc Schranz, and Larissa Schwaller (2025). “Using Natural Language Processing to Identify Monetary Policy Shocks”. Working Paper.

Tran, Paul L. (2026). “How Long Do Markets Need to Fully React to Monetary Policy Announcements?” Job Market Paper.

Figures and Tables

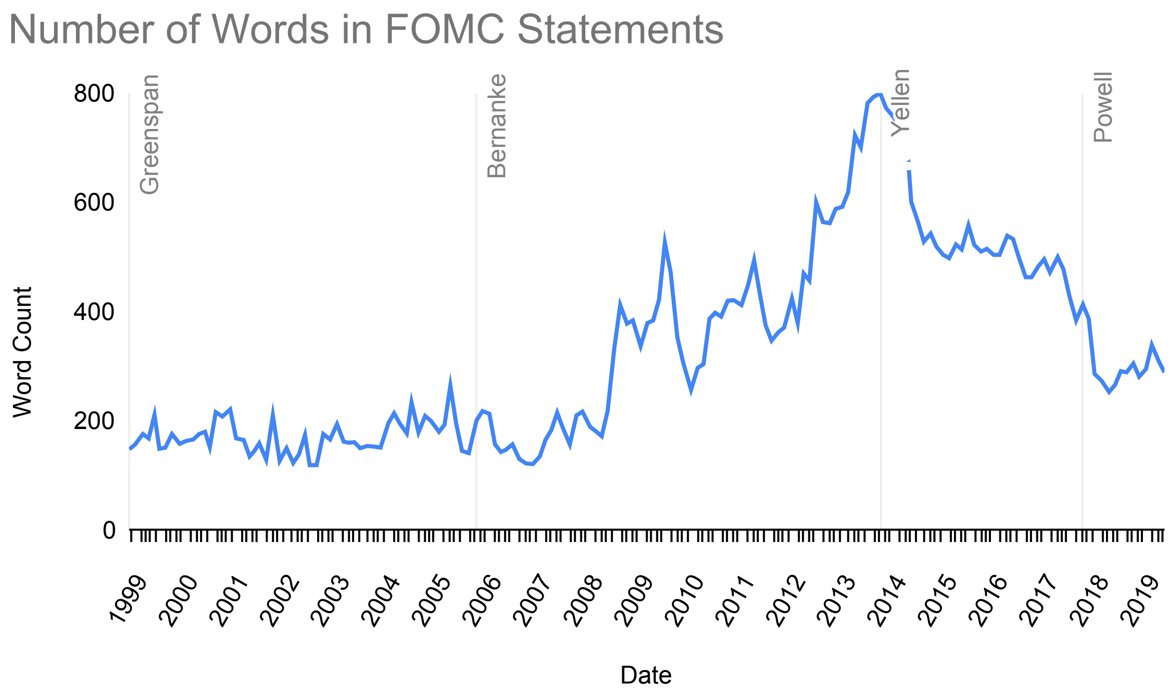

Figure 1: Number of Words in FOMC Statements, May 1999–October 2019

Notes: The above counts are for FOMC statements that have underwent pre-processing, which is explained in Subsection 2.1. From left to right, the vertical grey lines indicate the first FOMC meeting with Greenspan, Bernanke, Yellen, and Powell as Fed Chair.

Metric	$mp1_{\tau,t+40}$	$\|mp1_{\tau,t+40}\|$	$mp2_{\tau,t+40}$	$\|mp2_{\tau,t+40}\|$	Δr	$\|\Delta r\|$	S^1	D^1
Count	165	165	165	165	164	164	164	164
Mean	-0.004	0.022	-0.004	0.023	-0.018	0.118	0.751	0.249
SD	0.046	0.040	0.043	0.036	0.255	0.226	0.212	0.212
Max	0.125	0.280	0.170	0.185	0.500	1.250	0.984	0.800
75^{th}	0.005	0.0234	0.010	0.021	0	0.250	0.920	0.382
Median	0	0.006	0	0.010	0	0	0.826	0.174
25^{th}	-0.006	0	-0.011	0.005	0	0	0.622	0.079
Min	-0.280	0	-0.185	0	-1.250	0	0.200	0.016

Metric	$W_{RA,\tau}$	$W_{L,\tau}$	$W_{I,\tau}$	$W_{FC,\tau}$	$W_{FFR,\tau}$	$W_{BS,\tau}$	EC_{τ}	MP_{τ}
Count	165	165	165	165	165	165	165	165
Mean	0.267	0.103	0.239	0.056	0.255	0.080	0.665	0.335
SD	0.119	0.081	0.082	0.059	0.123	0.107	0.099	0.099
Max	0.681	0.295	0.421	0.228	0.693	0.368	0.898	0.693
75^{th}	0.361	0.169	0.290	0.072	0.313	0.152	0.728	0.397
Median	0.247	0.088	0.253	0.041	0.246	0	0.670	0.330
25^{th}	0.174	0.034	0.182	0	0.163	0	0.603	0.272
Min	0.041	0	0.078	0	0.068	0	0.307	0.102

Table 1: Descriptive Statistics

Notes: The expectations on federal funds rates for the current and next FOMC meetings are all calculated using high-frequency changes in federal funds futures prices within 50-minute windows surrounding the release of FOMC statements. Δr is the change in the target federal funds rate from the previous meeting. Variables between vertical bars represent absolute values. S^1 (D^1) is the cosine (dis)similarity measure between sequential FOMC statements. $W_{k,\tau}$ is the aggregated, statement-level topic share for topic k of FOMC statement date τ. EC_{τ} and MP_{τ} are thematic indices representing economic conditions and monetary policy tools, respectively, and build from the six k topic shares.

Topic Category	MSE	Pearson Correlation (ρ)
Real Activity	0.0182	0.9355
Labour Market	0.0065	0.9513
Inflation	0.0083	0.9707
Financial Conditions	0.0081	0.8540
Fed Funds Rate	0.0261	0.9008
Balance Sheet	0.0047	0.9743
Aggregate Metrics		
Mean	0.0120	0.9311
Median	0.0082	0.9434

Table 2: Model Validation Against Human Ground Truth

Notes: This table reports the Mean Squared Error (MSE) and Pearson correlation coefficients (ρ) between the probabilistic topic assignments of Gemini 3.1 Pro and the manual expert labels for a random subsample of 205 sentences (i.e., 10% of the corpus).

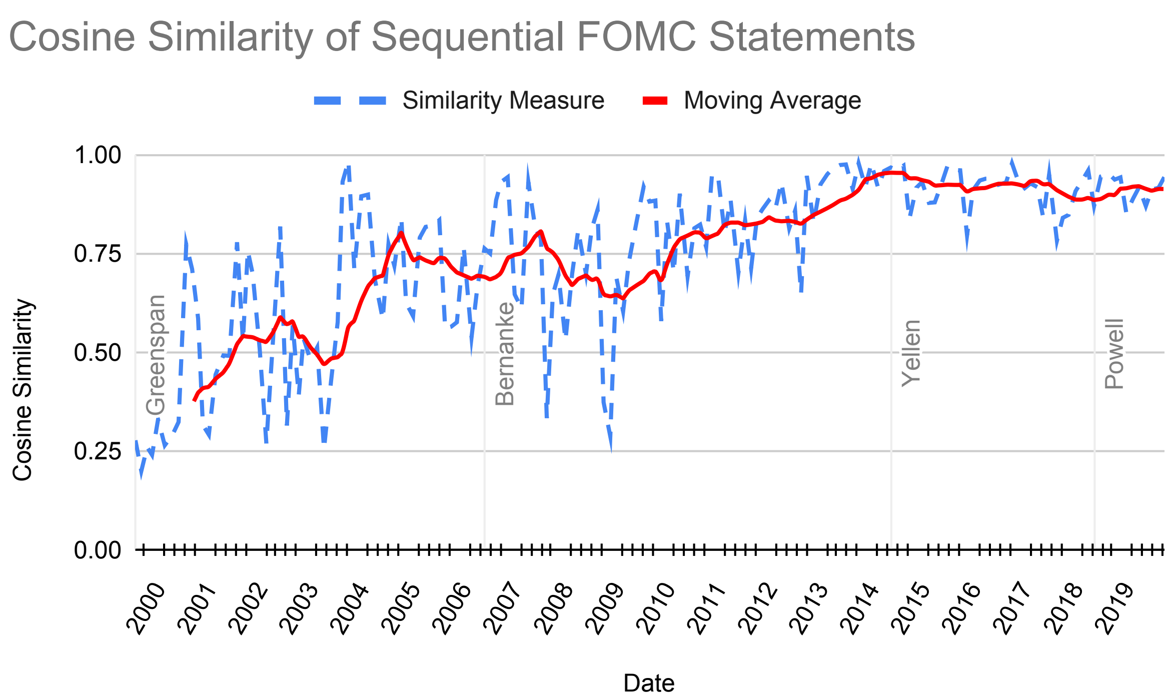

Figure 2: S^1 Cosine Similarity of Sequential FOMC Statements

Notes: S^1 is the cosine similarity between the TFIDF value of an FOMC statement and that of the immediately previous FOMC statement. Cosine similarity values closer to one (zero) mean the statements share more (less) common term usage. The moving average in solid red is calculated with a period of 10.

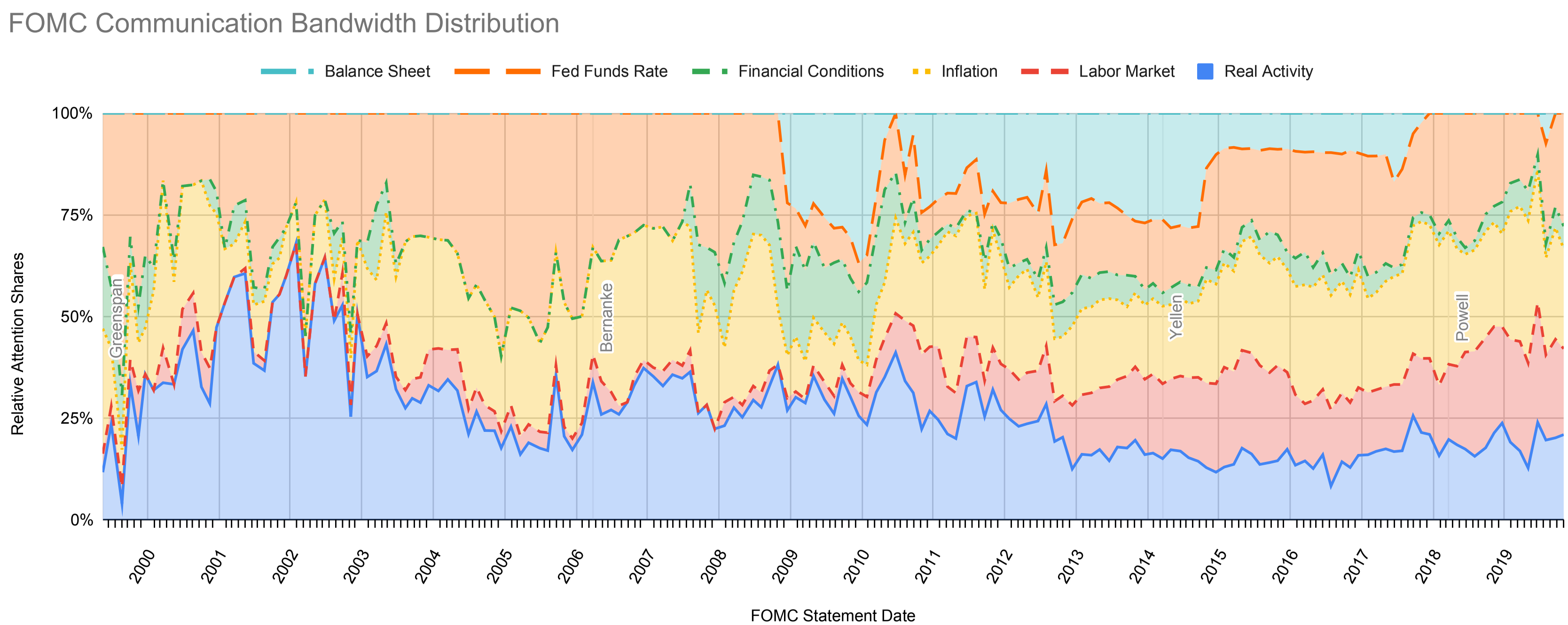

Figure 3: FOMC Statement Communication Bandwidth for May 1999–October 2019

Notes: This figure displays the statement-level topic shares, $W_{k,\tau}$, for the six thematic categories for May 1999–October 2019. Each shaded area represents the proportion of a statement's total word count dedicated to a specific topic, calculated as a word-weighted average of sentence-level probabilistic assignments from Gemini 3.1 Pro. Vertical dashed lines depict the tenures of Fed Chairs Greenspan, Bernanke, Yellen, and Powell. The six categories (Real Activity, Labour Market, Inflation, Financial Conditions, Fed Funds Rate, and Balance Sheet) are mutually exclusive and sum to 100%.

Topic Category	D^1	Δr
Real Activity	-0.030	-0.023
Labour Market	-0.051	0.001
Inflation	-0.084	0.193
Financial Conditions	-0.044	-0.021
Fed Funds Rate	0.056	0.046
Balance Sheet	0.124	-0.101

Table 3: Pearson Correlation Table

Notes: D^1 is semantic dissimilarity between sequential FOMC statements and Δr is the change in the target federal funds rate.

	$\|mp1_{\tau,t+40}\|$			$\|mp2_{\tau,t+40}\|$		
	(1)	(2)	(3)	(4)	(5)	(6)
ΔMP_τ	0.123***	0.105**	0.079**	0.097**	0.083**	0.050*
	(0.043)	(0.039)	(0.032)	(0.038)	(0.036)	(0.027)
D^1_τ		0.072***	0.044***		0.057***	0.021*
		(0.015)	(0.016)		(0.016)	(0.012)
$\tilde{I}_\tau$		0.130	0.053		0.125	0.027
		(0.143)	(0.136)		(0.131)	(0.122)
$\|\Delta r_\tau\|$			0.065***			0.082***
			(0.016)			(0.010)
Observations	164	164	164	164	164	164
R^2	0.079	0.224	0.325	0.061	0.175	0.378
Adj. R^2	0.073	0.209	0.308	0.055	0.160	0.363

Table 4: FOMC Statement Communication Bandwidth and MP Surprises

Notes: The dependent variables are the absolute changes in the current-meeting ($|mp1_{\tau,t+40}|$) and next-meeting ($|mp2_{\tau,t+40}|$) monetary policy surprises. Both surprises are measured within an event window starting 10 minutes before and ending 40 minutes after the FOMC statement release time. Columns (1) and (4) report Specification 1. Columns (2) and (5) report Specification 2. Columns (3) and (6) report Specification 3. ΔMP_τ represent the first-differenced thematic index for discussions about monetary policy tools. D^1_τ is semantic dissimilarity between sequential FOMC statements, $\tilde{I}_\tau$ is the interaction residual between D^1_τ and the thematic index for each regression specification. $|\Delta r_\tau|$ is the absolute change in the target federal funds rate. HAC standard errors are in parentheses. Constants included but not reported. * sig. at the 10% level. ** sig. at the 5% level. *** sig. at the 1% level.

A System Instructions for Gemini 3.1 Pro

The following is the set of system instructions provided to Google Gemini 3.1 Pro for its probabilistic topic classification task on each sentence within each FOMC statement. The six mutually exclusive topic categories that the LLM can assign probabilities to follow those from Dunn et al. (2024): Real Activity, Labour Market, Inflation, Financial Conditions, Fed Funds Rate, and Balance Sheet.

```
You are a macroeconomist who is part of an experiment.

TASK:
Analyze an ordered set of sentences from an FOMC statement. For each
↪ sentence, assign probability shares (0.0 to 1.0) across exactly SIX
↪ categories. The shares for each individual sentence MUST sum to
↪ exactly 1.0. The share values MUST be a float. DO NOT use strings or
↪ percentages.

THE SIX CATEGORIES (NOTE that the following terms and phrases are NOT
↪ exhaustive lists for each category):
1. Real Activity: Gross domestic product (GDP), economic growth, output,
↪ household spending, consumer spending, business fixed investment,
↪ residential investment, industrial production, manufacturing,
↪ exports, imports, inventory, consumption.
2. Labor Market: Maximum employment, payrolls, job gains, unemployment
↪ rate, labor force participation, labor supply, vacancies, hiring,
↪ labor market tightening, quit rate, layoffs.
3. Inflation: Personal Consumption Expenditures (PCE) price index,
↪ Consumer Price Index (CPI), price stability, inflation expectations,
↪ core inflation, symmetric target, headline inflation, energy prices,
↪ food prices, price pressures.
4. Financial Conditions: Systemic risk, banking system, leverage,
↪ vulnerabilities, capital ratios, liquidity, financial system
↪ resilience, credit standards, stress, financial cycle, banking
↪ sector, credit spreads, equity prices, stock market, Treasury
↪ yields, international developments, global conditions, exchange
↪ rates, foreign economies, market volatility, financial markets.
5. Fed Funds Rate: Target range, policy rate, federal funds rate, basis
↪ points, liftoff, accommodation, policy stance, effective federal
↪ funds rate, restrictive stance, neutral rate.
6. Balance Sheet: System Open Market Account (SOMA), asset purchases,
↪ Quantitative Easing (QE), Quantitative Tightening (QT),
↪ reinvestment, principal payments, mortgage-backed securities (MBS),
↪ Treasuries, balance sheet normalization, holdings, principal
↪ payments

CORE LOGIC:
```

```
- LATENT KNOWLEDGE: Use your specialized knowledge of FOMC jargon to
↪  identify synonyms.
- CONTEXTUAL INTEGRITY: Use the entire statement provided as context to
↪  resolve pronouns or references. DO NOT USE EXTERNAL INFORMATION OR
↪  KNOWLEDGE OF FUTURE EVENTS.
- SYNTACTIC WEIGHTING: Use your internal training on English grammar and
↪  syntax to identify the primary anchor of the sentence. Assign a
↪  majority weight (0.50 to 0.70) to the primary grammatical subject,
↪  the active policy instrument, or the main clause.
  * ANCHOR HEURISTIC: In structures where a policy stance or action is
   ↪  the "means" for achieving outcomes (e.g., "With [A], [B] will
   ↪  occur" or "[A] to support [B]"), prioritize [A].
  * CONCESSIVE HEURISTIC: In structures involving qualifiers (e.g., "[A]
   ↪  despite [B]" or "[A] though [B]"), prioritize the primary
   ↪  assertion [A].
  * NOTE: These heuristics are non-exhaustive examples. Apply your
   ↪  understanding of syntactic primacy to all sentence structures to
   ↪  ensure the "driver" or "news" of the sentence receives the highest
   ↪  share.
- CONTEXTUAL PROPAGATION: For sentences that describe the Committee
↪  "monitoring," "anticipating," or "evaluating" the situation without
↪  naming specific economic variables, you MUST distribute the
↪  probability shares based on the primary economic anchors identified
↪  in the immediately preceding sentences.

OUTPUT FORMAT:
Return ONLY a JSON list of objects. No prose.
```

B Human Validation Guidebook and Coding Rules

The following is the guidebook and specific set of coding rules I follow to manually assign topic shares to each sentence of the randomly selected 10% subsample of my corpus of FOMC statements. These rules are developed based on macroeconomic theory and the grammatical structure of English and allow me to assign nuanced, non-binary topic shares that can have unequal weighting (e.g., 1.0, 0.5/0.5, or 0.7/0.3) when a sentence conveys multiple policy signals. The six mutually exclusive topic categories follow those from Dunn et al. (2024): Real Activity, Labour Market, Inflation, Financial Conditions, Fed Funds Rate, and Balance Sheet.

```
The topics are: Real Activity, Labor Market, Inflation, Financial
↪  Conditions, Fed Funds Rate, Balance Sheet. Below is a non-exhaustive
↪  list of "anchor words" I find and associate with each topic:
```

```
1. Real Activity: Gross domestic product (GDP), economic growth, output,
 ↪  household spending, consumer spending, business fixed investment,
 ↪  residential investment, industrial production, manufacturing,
 ↪  exports, imports, inventory, consumption.
2. Labor Market: Maximum employment, payrolls, job gains, unemployment
 ↪  rate, labor force participation, labor supply, vacancies, hiring,
 ↪  labor market tightening, quit rate, layoffs.
3. Inflation: Personal Consumption Expenditures (PCE) price index,
 ↪  Consumer Price Index (CPI), price stability, inflation expectations,
 ↪  core inflation, symmetric target, headline inflation, energy prices,
 ↪  food prices, price pressures.
4. Financial Conditions: Systemic risk, banking system, leverage,
 ↪  vulnerabilities, capital ratios, liquidity, financial system
 ↪  resilience, credit standards, stress, financial cycle, banking
 ↪  sector, credit spreads, equity prices, stock market, Treasury
 ↪  yields, international developments, global conditions, exchange
 ↪  rates, foreign economies, market volatility, financial markets.
5. Fed Funds Rate: Target range, policy rate, federal funds rate, basis
 ↪  points, liftoff, accommodation, policy stance, effective federal
 ↪  funds rate, restrictive stance, neutral rate.
6. Balance Sheet: System Open Market Account (SOMA), asset purchases,
 ↪  Quantitative Easing (QE), Quantitative Tightening (QT),
 ↪  reinvestment, principal payments, mortgage-backed securities (MBS),
 ↪  Treasuries, balance sheet normalization, holdings, principal
 ↪  payments

In order to manually assign probability shares of topics for each
 ↪  sentence, I first read the entire cleaned FOMC statement that said
 ↪  sentence belongs to. This reading is done because I use the entire
 ↪  statement as context  for assigning topic probability shares for
 ↪  each sentence.

The following are the rules that I follow to the best of my ability in
 ↪  order to assign probability shares for topics:
Rule 1, Case A: A sentence is assigned a weight of 100% (1) to a single
 ↪  topic if one or more anchor words associated with said topic is
 ↪  found in the sentence, and no other anchor words of any other topic
 ↪  is found.
Rule 1, Case B: If no explicit anchor words are found in the sentence, I
 ↪  read the immediate context (i.e., previous and immediate sentences)
 ↪  within the same paragraph to determine the indirect/latent topic.
Rule 1, Case C: If no explicit anchor words are found in the sentence
 ↪  and the sentence describes the general voting procedure/process,
 ↪  meeting schedule, or "stance of policy", my default assignment is
 ↪  100% to the topic of "Fed Funds Rate".
```

```
Rule 2, Case A: If two distinct topics are mentioned in a sentence, each
↪  are grammatically full clauses by themselves, and are joined by an
↪  "equaliser" word (e.g., and, but, while, or, as well as), then each
↪  topic is assigned a probability share of 50% (0.5).
Rule 2, Case B: If more than two (i.e., N) distinct topics are mentioned
↪  in a sentence, each are grammatically full clauses by themselves,
↪  and are joined by "equaliser" words, then each topic is assigned a
↪  probability share of 1/N.
Rule 2, Case C: If the FOMC is highlighting a difference between
↪  distinct topics in a sentence (i.e., contrastive), but each topic is
↪  contained in a full clause and is the subject, they receive equal
↪  probability shares.
Rule 2, Case D: If a clause within a sentence is a list of distinct
↪  topics, then the probability share assigned to that entire clause
↪  must be split evenly or accordingly to each topic discussed within
↪  that sentence clause.
Rule 3, Case A: If a sentence identifies the primary topic being
↪  monitored or discussed for its effect on a secondary topic, the
↪  primary topic is assigned 70% (0.7) and the secondary topic 30%
↪  (0.3).
Rule 3, Case B: If a sentence identifies the primary topic, following by
↪  a phrase that is introduced by phrases such as "including,
↪  reflecting, such as" which itself contains distinct topics, 0.7 is
↪  assigned to the primary topic and 0.3 is evenly distributed to the
↪  secondary topics.
Rule 3 Note: A 70/30 split was chosen to acknowledge the presence of
↪  secondary economic topics in a sentence without assigning to it
↪  equal probability share to the primary topic/grammatical subject of
↪  the sentence.
Rule 4: When unsure about a grammatical structure (e.g., whether a
↪  phrase constitutes an independent clause or a dependent modifier),
↪  the researcher shall not provide the specific sentence text to the
↪  LLM. Instead, the researcher will provide a Syntactic Template using
↪  placeholders (X, Y, Z) to the LLM when asking for clarification.
↪  This abstraction prevents any knowledge leaks to the LLM.
```

Appendix Figures

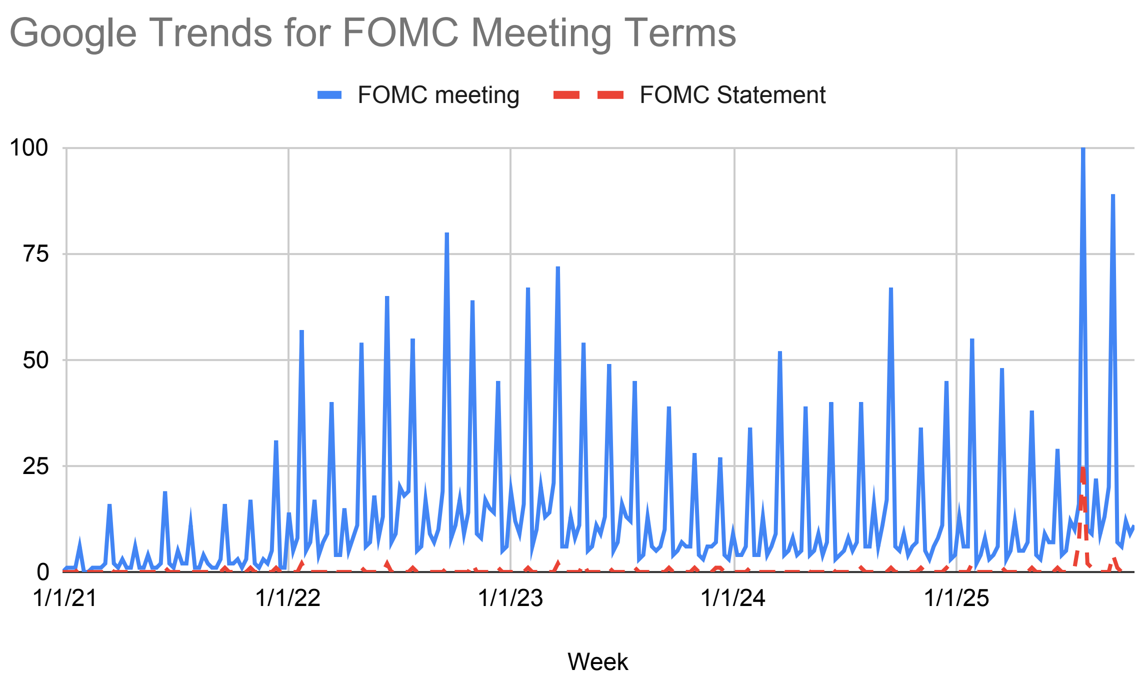

Figure C1: Google Trends Interest About FOMC Statements over Time, January 2021–October 2025

Notes: The blue and solid (red and dashed) line represents the search interest for the phrase "FOMC meeting" ("FOMC statement") in the U.S.

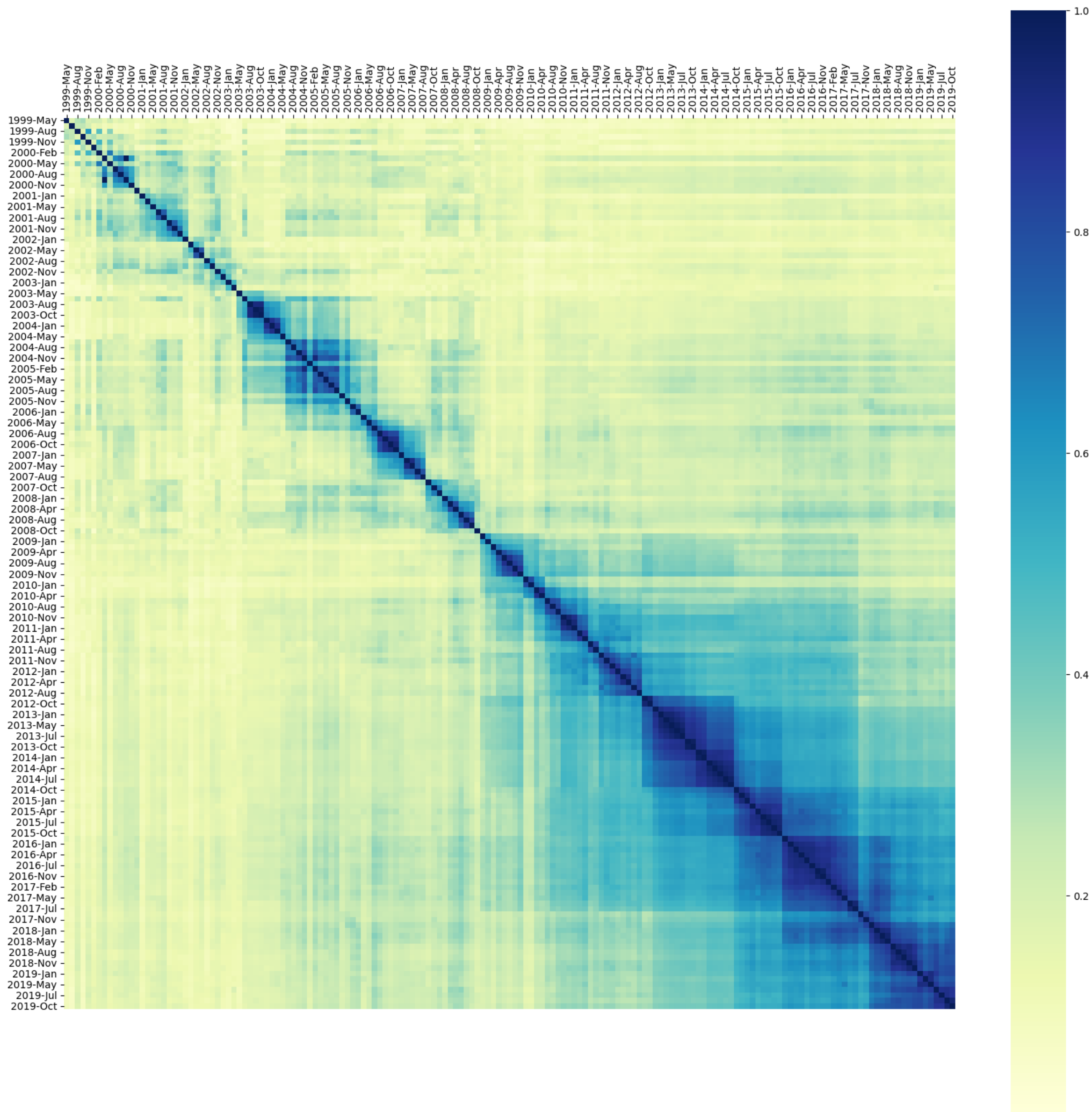

Figure C2: Document Similarity Matrix for FOMC Statements

Notes: Each element is the cosine similarity between two FOMC statements of that row and column. The cosine similarity value measures how similar the terms of both statements are. The darker shade of blue represents pairs of documents that have higher similarity measures closer to one, while the light shade of green represents pairs of documents that do not have terms in common and similarity values closer to zero. The main diagonal has all ones because the cosine similarity value is calculated between an FOMC statement with itself. All FOMC statements have been preprocessed such that all words are lowercase, all words are reduced to base form, and all stop-words are removed.

Appendix Tables

	$\lvert mp1_{\tau,t+40}\rvert$			$\lvert mp2_{\tau,t+40}\rvert$		
	(1)	(2)	(3)	(4)	(5)	(6)
$\lvert\Delta r_\tau\rvert$	0.089***		0.072***	0.095***		0.086***
	(0.014)		(0.016)	(0.011)		(0.011)
D^1_τ		0.077***	0.044***		0.095***	0.021*
		(0.017)	(0.016)		(0.018)	(0.012)
Observations	164	164	164	164	164	164
R^2	0.249	0.163	0.293	0.349	0.126	0.361
Adj. R^2	0.245	0.158	0.285	0.345	0.121	0.354

Table 5: Regression Results without FOMC Statement Communication Bandwidth
Notes: The dependent variables are the absolute changes in the current-meeting ($\lvert mp1_{\tau,t+40}\rvert$) and next-meeting ($\lvert mp2_{\tau,t+40}\rvert$) monetary policy surprises. Both surprises are measured within an event window starting 10 minutes before and ending 40 minutes after the FOMC statement release time. $\lvert\Delta r_\tau\rvert$ is the absolute change in the target federal funds rate. D^1_τ is semantic dissimilarity between sequential FOMC statements. HAC standard errors are reported in parentheses. Constants are included in all specifications but are not reported. * sig. at the 10% level. ** sig. at the 5% level. *** sig. at the 1% level.